

1a	Vertically upward arrow from point B representing upthrust, and vertically downward arrow from point C representing weight of tube and sand. Both arrows are equal in length, representing equal magnitude.
1b	Upthrust for both cases are the same, equal to weight of tube and sand. Upthrust = weight of fluid displaced $\rho_w A L_w g = \rho_l A L_l g = \text{weight of tube and sand}$ $L_l = \rho_w L_w / \rho_l$ $= 0.99 \times 12.1 / 1.11$ $= 10.792 \text{ cm}$ Change in L = 10.792 – 12.1 $= -1.3 \text{ cm}$
1ci	Its weight and upthrust provide couple / restoring torque to keep tube upright.
1cii	When the tube is displaced downward, the <u>upthrust is greater than the tube's weight</u>. Hence, the net force on the block is <u>upward</u>, opposite to the displacement. When the block is displaced upward, the <u>block's weight is greater than the upthrust</u>. Hence, the net force on the block is <u>downward</u>, opposite to the displacement.
2a	$g = GM/R^2$, so if distances near surface of planet is much smaller than R, g is a constant